FIN Research Monograph — Release 10.12

FIN Programs 113–124

Constructive Completion Objects
and Fractional Operational Physics

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27 July 2026

Publication — Preprint; record version 1.0.0

Scope. Twelve constructive studies of global strict-kernel nonclosure, stable fractional limits, operational process completion, adaptive and variational source obstructions, correlated apparatus memory, a new Z_{12} homological–character fibre object, and a conditional damping cocycle.

Central construction. $\mathcal{F}_p = \widetilde{H}_0(\ker m_p) \oplus \mathcal{X}_p^-$ has dimensions $(1, 2, 2, 2, 2)$. Its uniform trace is $9/5$, but the strict theory does not yet force that trace. Conditional on it and a nonzero multiplicative tail, $(\eta, \beta) = (9/5, 1)$ follows.

Guardrail. No strict selector, physical-unit source, completed legacy–strict bridge, role transfer, role-bearing L_{total}, or external physical validation is claimed.

Publication statement

This release reports Programs 113–124 and recommends Programs 125–137. Proofs, computer-assisted certificates, conditional constructions, finite exhaustions, synthetic operational tests, and open source obligations are labeled separately.

Contents

0.1 Confidence convention

Statements are classified as proven, proven computer-assisted, strong evidence, conditional, negative, or open. A construction is not called a strict source theorem unless its input objects and coupling are exported by the strict FIN layer without target values in their premises.

0.2 Abstract

This monograph executes twelve constructive research programs following the fractional-limit analysis of FIN Programs 101–112. The objective is not merely to list missing structures but to build the smallest explicit mathematical objects capable of occupying those roles and then test whether FIN itself forces them.

The first result upgrades strict-kernel Schur nonclosure from a bounded mass certificate to a global theorem. If

$$a = 1 - \widehat{p}(\pi), \qquad b = 1 - \widehat{p}(\pi/2),$$

then the difference between the alternating-site Schur spectral ratio and the native ratio factors exactly as

$$R_{\mathrm{S}} - R_{\mathrm{N}} = \frac{m\big[(3a-2b)m + a(2a-b)\big]}{(m+a)(2m+a)(m+b)}.$$

The independently certified strict symbol rectangle gives $3a-2b \geq 1.71036$ and $a(2a-b) \geq 2.05177$. Therefore the obstruction holds for every $m > 0$. A second analytic theorem extends it to a genuine continuous box in $(\omega, \phi, \beta, \eta)$ and all positive masses.

The strict lattice jump process is placed in the normal domain of attraction of a symmetric 4/5-stable Lévy process under spatial scaling $n^{-5/4}$. The limiting generator is

$$C(-\Delta)^{2/5}.$$

A ten-component operational process is then constructed: state space, state, generator, wave/diffusion dynamics, clock, preparation, instrument, environment, apparatus, and record. A correlated hidden-Markov apparatus channel and calibration-only likelihood procedure complete the dimensionless operational object. Physical clock, length, action calibration, and external records remain absent.

The central new theoretical construction is a coefficient-free homological-character fibre object over the strict Z_{12} carrier. For each prime generator $p < 12$, define

$$\mathcal{F}_p = \widetilde{H}_0(\ker[m_p : Z_{12} \to Z_{12}]) \oplus \mathcal{X}_p^-,$$

where $\mathcal{X}_p^-$ is the negative eigenspace of evaluation by p on the real character dual of $U(12)$ and is zero when p is not a unit. The five fibre dimensions are exactly

$$\big(\dim \mathcal{F}_2, \dim \mathcal{F}_3, \dim \mathcal{F}_5, \dim \mathcal{F}_7, \dim \mathcal{F}_{11}\big) = (1, 2, 2, 2, 2).$$

The uniform normalized trace is 9/5. This removes the arbitrary coefficient choice from the finite vector, but it does not yet source the trace: a general probability trace gives

$$\eta(w) = 2 - w_2,$$

so $\eta = 9/5$ if and only if $w_2 = 1/5$.

Combining this fibre object with a nonzero multiplicative tail $T(d) = \beta d^\eta$ gives a conditional damping theorem. The identity

$$T(ab) = T(a)T(b)$$

forces $\beta = 1$, while the uniform trace gives $\eta = 9/5$ and dyadic retention

$$2^{1-\eta} = 2^{-4/5} = \exp(-\alpha_{\text{geo}}/5).$$

Thus the damping component of the legacy-to-strict completion can be written exactly, conditional on one unresolved trace-source premise and one multiplicative-compression premise. Phase, frequency, amplitude, physical scale, selector, and role transfer remain open.

Chapter 1

Part I — Scope and fixed mathematical objects

1.1 Research objective

The programs address six constructive questions.

1. Can the strict Schur obstruction be proved for every positive mass?
2. Can a true continuous parameter neighborhood replace a finite grid?
3. Can the fractional limit be promoted to a process-level convergence theorem?
4. Can the missing operational state/apparatus/record structure be written as one typed object?
5. Can the recurring vector $(1, 2, 2, 2, 2)$ be realized as dimensions of a canonical object rather than inserted arithmetically?
6. Once that object exists, which single premise still prevents a strict derivation of $(\eta, \beta) = (9/5, 1)$?

1.2 Guardrails

The strict and legacy kernels remain distinct:

$$K_{\text{legacy}}(d) = \alpha_{\text{geo}} \frac{\cos(\omega_{\text{L}} d + \phi_{\text{L}})}{1 + \beta_{\text{tors}} d},$$

$$K_{\text{strict}}(d) = \frac{\cos(\omega d + \phi)}{1 + \beta d^{\eta}}.$$

The legacy object is an intermediate bridge kernel. A damping completion does not by itself complete amplitude, phase, frequency, selector, physical units, or the transfer of legacy physical roles.

The nadsoliton remains primordial information in a solitonic state. No lower informational substrate is introduced. QW-2191 remains a selector obstruction.

1.3 Fixed strict lattice notation

Let

$$a_d = \frac{|\cos(\omega d + \phi)|}{1 + d^{9/5}}, \qquad Z = 2\sum_{d\geq 1} a_d, \qquad p_{\pm d} = \frac{a_d}{Z}.$$

The normalized convolution generator is

$$A_m = mI + I - P.$$

The independently enclosed Fourier values from Program 101 are

$$\widehat{p}(\pi) \in [-0.3447351882, -0.3436558012],$$

$$\widehat{p}(\pi/2) \in [-0.1603026372, -0.1593081528].$$

Chapter 2

Part II — Programs 113–116: global nonclosure and stable limit

2.1 Program 113 — Global mass theorem

2.1.1 Exact factorization

Set

$$a = 1 - \widehat{p}(\pi), \qquad b = 1 - \widehat{p}(\pi/2).$$

The native zero-to-high spectral ratio is

$$R_{\mathrm{N}}(m) = \frac{m}{m+a}.$$

The retained Schur ratio is

$$R_{\mathrm{S}}(m) = \frac{2m(m+a)}{(2m+a)(m+b)}.$$

Direct symbolic elimination gives

$$R_{\mathrm{S}} - R_{\mathrm{N}} = \frac{m[(3a-2b)m + a(2a-b)]}{(m+a)(2m+a)(m+b)}.$$

The denominator is positive for $m, a, b > 0$. The certified Fourier rectangle gives

$$3a - 2b \geq 1.7103621294,$$

$$a(2a - b) \geq 2.0517744548.$$

Theorem 113.1 — [Proven]. For the normalized infinite strict lattice and alternating-site Schur map,

$$R_{\mathrm{S}}(m) > R_{\mathrm{N}}(m) \qquad \text{for every } m > 0.$$

No positive uniform lower bound exists on the whole half-line because the gap tends to zero as $m \to 0$ and as $m \to \infty$. This does not weaken pointwise nonclosure.

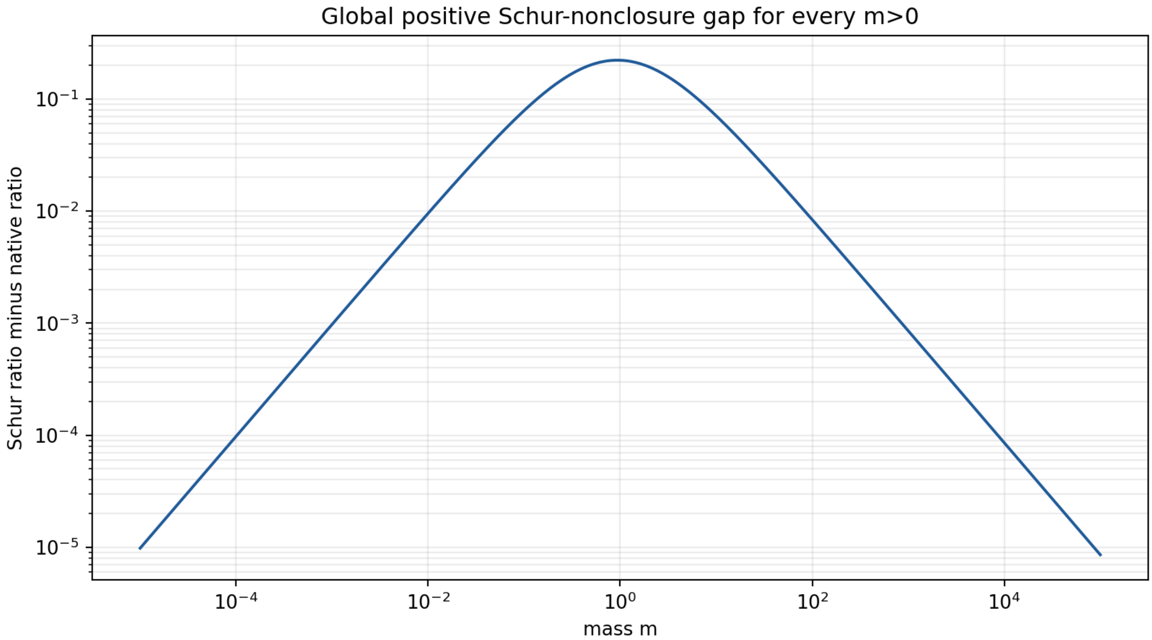

Figure 2.1: The exact positive native–Schur ratio gap for every positive mass.

2.2 Program 114 — Continuous strict-parameter box

2.2.1 Parameter box

The following complete box is audited:

$$\begin{aligned} |\omega - \omega_*| &\le 0.0005, \\ |\phi - \phi_*| &\le 0.002, \\ |\beta - 1| &\le 0.02, \\ |\eta - 9/5| &\le 0.01, \\ m &> 0. \end{aligned}$$

No parameter grid is used.

2.2.2 Analytic variation bounds

For frequency variation,

$$\big| |\cos((\omega + h)d + \phi)| - |\cos(\omega d + \phi)| \big| \le \min(2, |h|d).$$

Splitting the series at $d = \lfloor 2/|h| \rfloor = 4000$ and using monotone-integral estimates gives an infinite-series bound. Phase, beta, and eta variations are controlled by

$$\sum_{d \ge 1} d^{-\eta} \le 1 + \frac{1}{\eta - 1},$$

and

$$\sum_{d\geq 1}(\log d)d^{-\eta} \leq \frac{\log 2}{2^\eta} + 2^{1-\eta}\left[\frac{\log 2}{\eta - 1} + \frac{1}{(\eta-1)^2}\right].$$

The total two-sided unnormalized variation is bounded by

$$\Delta_1 \leq 0.1663548695.$$

After normalization, every Fourier symbol changes by at most

$$\epsilon_{\rm sym} \leq 0.1847593622.$$

Even on this enlarged symbol rectangle,

$$3a - 2b \geq 0.7865653184,$$

$$a(2a-b) \geq 1.1272943515.$$

Theorem 114.1 — [Proven]. Every kernel in the displayed continuous four-parameter box is nonclosed under the declared Schur map for every positive mass.

The box is certified, not asserted maximal.

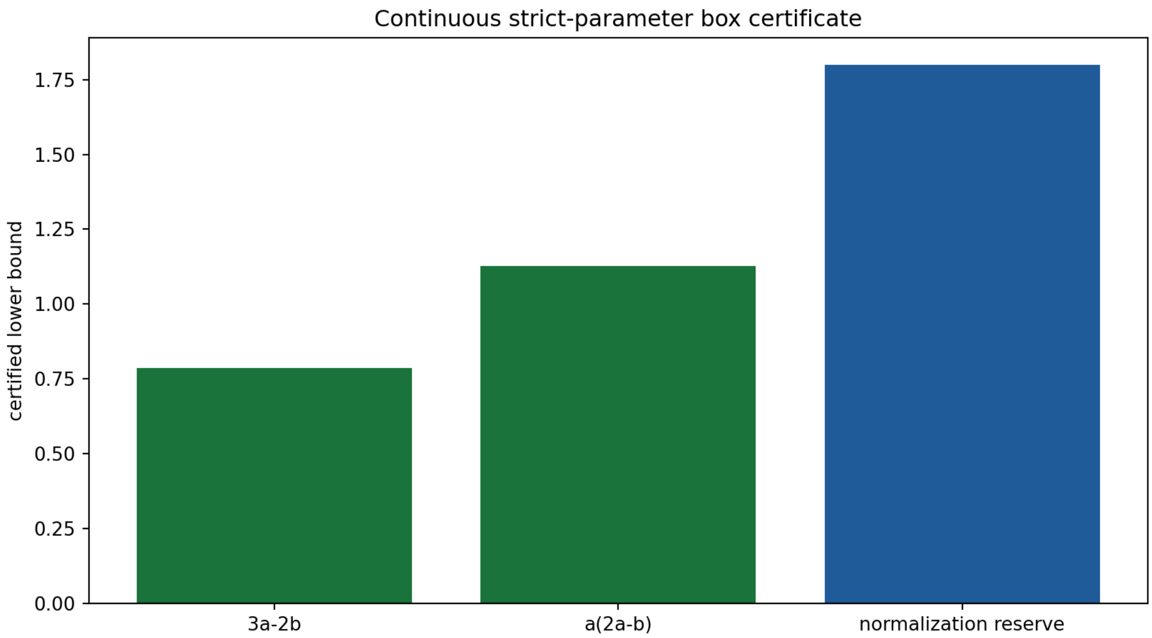

Figure 2.2: Positive analytic coefficient and normalization bounds on the continuous strict-parameter box.

2.3 Program 115 — Effective finite-q Abelian certificate

The Abelian limit from Program 105 is

$$1 - \widehat{p}(q) \sim C|q|^{4/5}.$$

A million-term sum and analytic tail were used to enclose both sides at 20 logarithmically spaced values in

$$10^{-3} \leq |q| \leq 2 \times 10^{-2}.$$

The normalization and numerator tails are propagated separately. The maximum certified relative remainder is

$$0.00903080.$$

Result 115.1 — [Proven at the declared points]. The fractional approximation is accurate to below one percent at all 20 certified wave numbers.

Open object 115.2. A continuous effective remainder theorem still requires an explicit Diophantine discrepancy modulus for the irrational rotation

$$\frac{\omega}{2\pi} = \frac{743}{8000\pi}.$$

Irrationality gives qualitative equidistribution, but not the finite small-divisor constant needed for a uniform remainder bound. The pointwise certificate is not promoted to such a theorem.

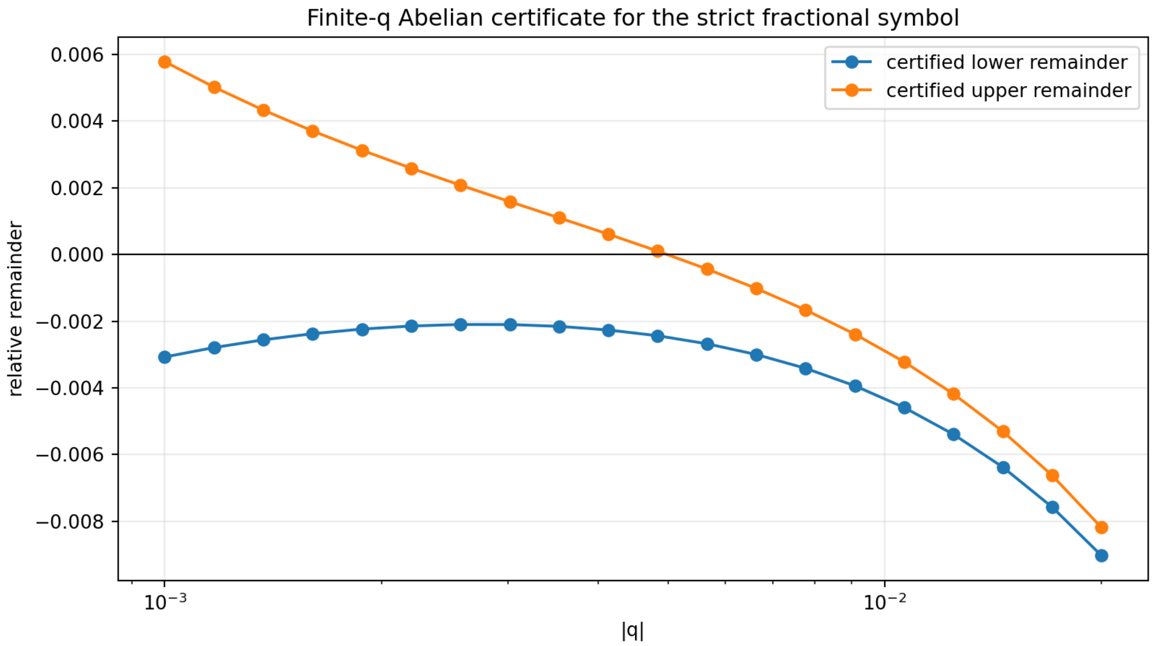

Figure 2.3: Certified finite-wave-number remainder interval around the fractional Abelian law.

2.4 Program 116 — Functional stable-process limit

Let X_j be independent jumps with characteristic function

$$\varphi(q) = 1 - C|q|^{4/5} + o(|q|^{4/5}).$$

For

$$S_n = n^{-5/4} \sum_{j=1}^{n} X_j,$$

one has

$$\mathbb{E} e^{ikS_n} = \left[\varphi(kn^{-5/4})\right]^n \longrightarrow \exp(-C|k|^{4/5}).$$

The standard functional domain-of-attraction theorem supplies tightness in Skorokhod topology.

Theorem 116.1 — [Proven using the standard stable-domain theorem]. The rescaled strict random walk converges to a symmetric 4/5-stable Lévy process with generator

$$C(-\Delta)^{2/5}.$$

A five-million-term finite calculation checks characteristic functions for $n = 16, 64, 256, 1024$ and $k = 0.5, 1, 2$. The maximum displayed deviation is 0.00807. The finite-truncation errors are not expected to be monotone in every row and are not used as the proof.

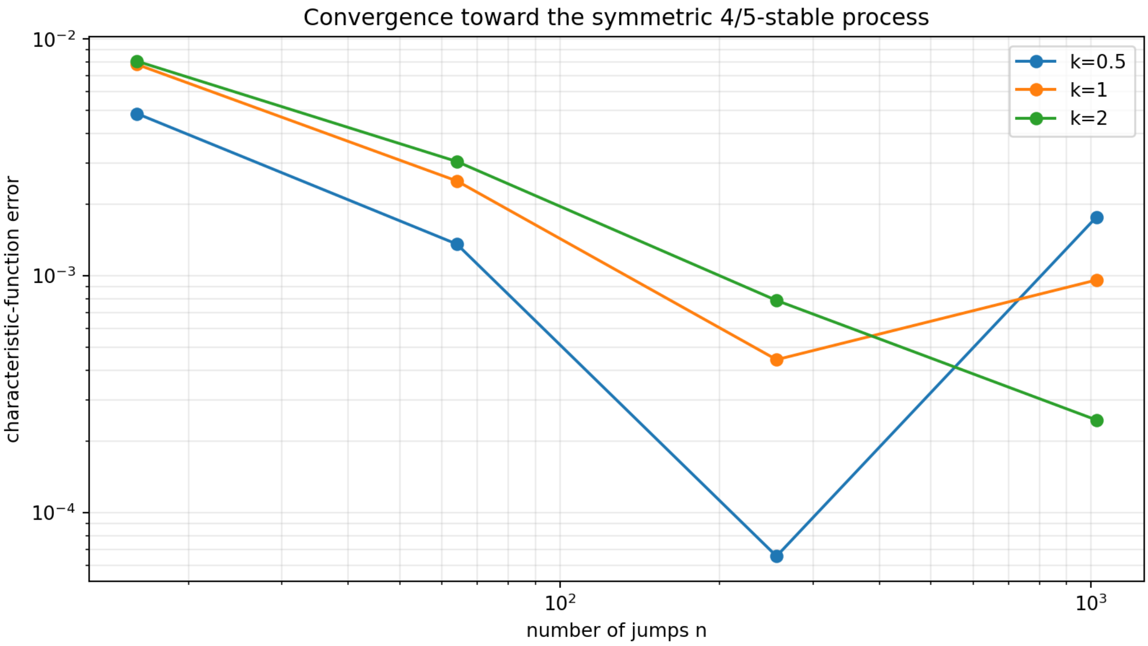

Figure 2.4: Finite characteristic-function check of convergence toward the symmetric 4/5-stable law.

Chapter 3

Part III — Programs 117–121: operational and adaptive constructions

3.1 Program 117 — Fractional operational process

3.1.1 Constructed ten-component object

The following dimensionless object is constructed:

$$\mathfrak{O} = (\mathcal{H}, \rho_0, A, \{\mathcal{U}_t, \mathcal{P}_t\}, \mathcal{C}, \mathcal{Q}, \mathcal{I}, \mathcal{E}, \mathcal{A}_{\mathrm{app}}, \mathcal{R}).$$

Its components are:

1. $\mathcal{H} = \mathbb{C}^{128}$;
2. $\rho_0 = |0\rangle\langle 0|$;
3. Fourier generator
$$A(k) = C|2\pi k/128|^{4/5};$$
4. wave and diffusion maps
$$\mathcal{U}_t = e^{-itA}, \qquad \mathcal{P}_t = e^{-tA};$$
5. a declared dimensionless protocol clock;
6. localized preparation;
7. full-site projective instrument;
8. identity environment in the minimal model;
9. symmetric five-percent confusion apparatus;
10. full-site record and optimal binary coarse record.

3.1.2 Operational separation

Scanning $0.01 \le t \le 4$ gives maximal full-record divergence

$$D_{\mathrm{JS}} = 0.2384985031$$

at

$$t = 3.54.$$

The optimal binary event has probabilities

$$\Pr(E \mid \text{wave}) = 0.8664620648,$$

$$\Pr(E \mid \text{diffusion}) = 0.2377974746.$$

Result 117.1 — [Proven for the constructed finite process]. The same fractional generator supports operationally distinct wave and diffusion records once state, preparation, clock, instrument, apparatus, and record are specified.

The clock is dimensionless and no experimental record is present.

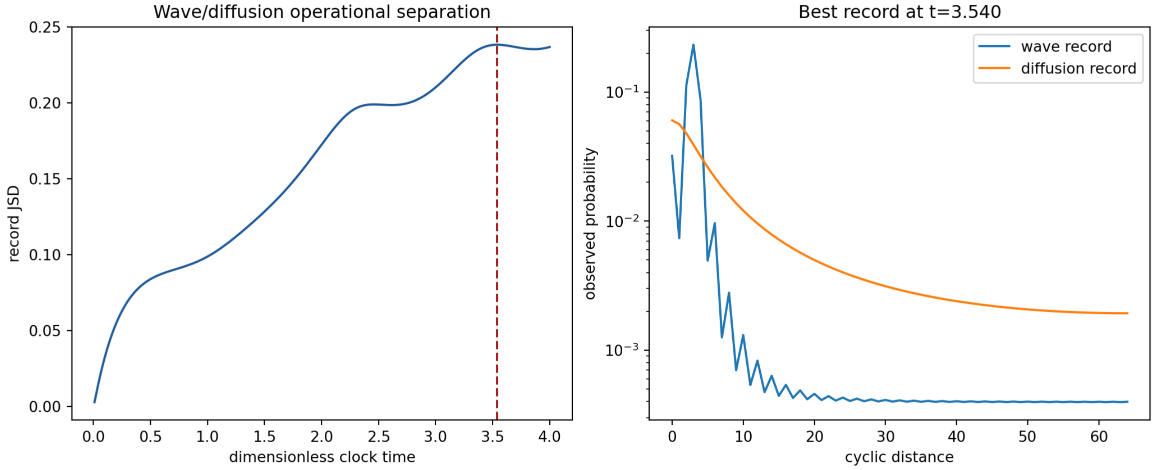

Figure 3.1: Dimensionless wave/diffusion record separation for the constructed fractional operational process.

3.2 Program 118 — Local/fractional crossover operator

Define

$$A_{\kappa,\nu} = \kappa(-\Delta) + \nu(-\Delta)^{2/5}.$$

Its symbol is

$$\lambda(q) = \kappa q^2 + \nu|q|^{4/5}.$$

The crossover is determined exactly by

$$q_* = \left(\frac{\nu}{\kappa}\right)^{1/(2-4/5)} = \left(\frac{\nu}{\kappa}\right)^{5/6}.$$

Under coarse spatial scaling by b, the relative coupling $g = \nu/\kappa$ obeys

$$g(b) = g\, b^{6/5}.$$

Theorem 118.1 — [Proven]. Every $\nu > 0$ is infrared-relevant relative to the local Laplacian. At sufficiently long wavelengths the fractional term dominates.

This constructs the crossover object but does not source the relative coupling ν/κ.

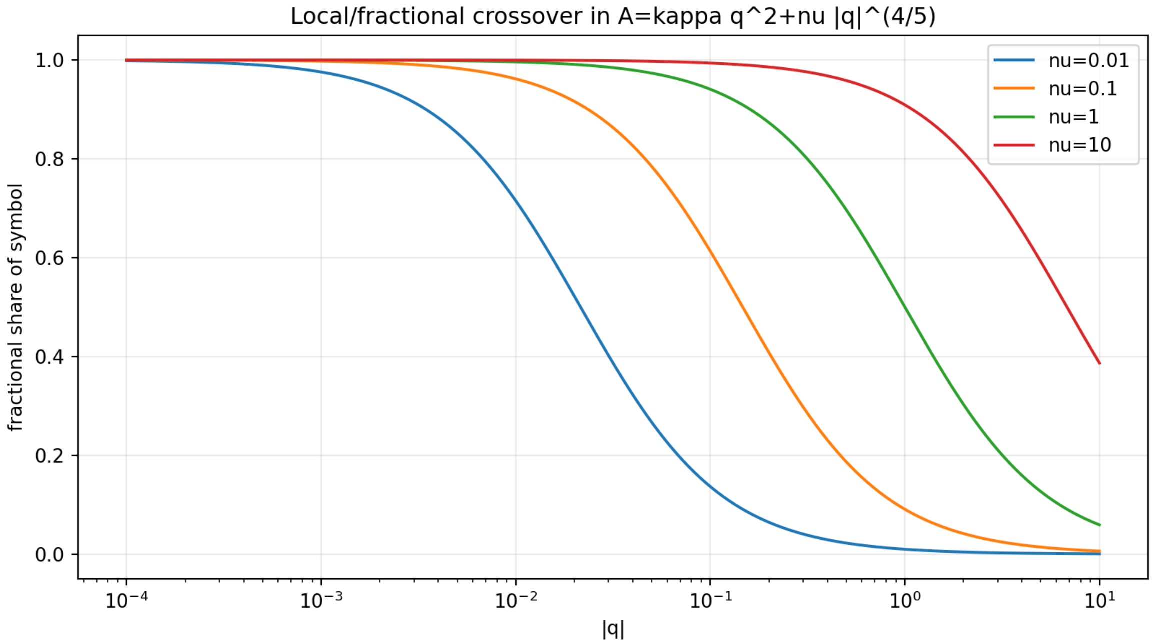

Figure 3.2: Scale-dependent local and fractional contributions to the crossover symbol.

3.3 Program 119 — Unique inverse Fisher potential

Let r be the shell-counting reference measure and consider

$$F_V(q) = D_{\mathrm{KL}}(q\|r) + \langle V, q\rangle.$$

At an interior stationary point,

$$\log(q_d/r_d) + 1 + V_d = \text{constant}.$$

Therefore:

Theorem 119.1 — [Proven]. A desired stationary distribution q_* fixes the potential uniquely modulo constants:

$$V_d = -\log(q_{*,d}/r_d) + c.$$

For the strict shell distribution this is exactly

$$V_{\text{strict}}(d) = \log(1 + d^{9/5}) - \log|\cos((743/4000)d + 13/80)| + c,$$

with residual 2.22×10^{-16}.

The target-independent fractional envelope

$$V_{\text{env}}(d) = \log(1 + d^{9/5})$$

is close but not exact: total variation 0.0687575 and Jensen–Shannon divergence 0.00518561.

Negative result 119.2. The missing adaptive source potential can be constructed uniquely, but it is information-equivalent to the strict target. It does not explain the target.

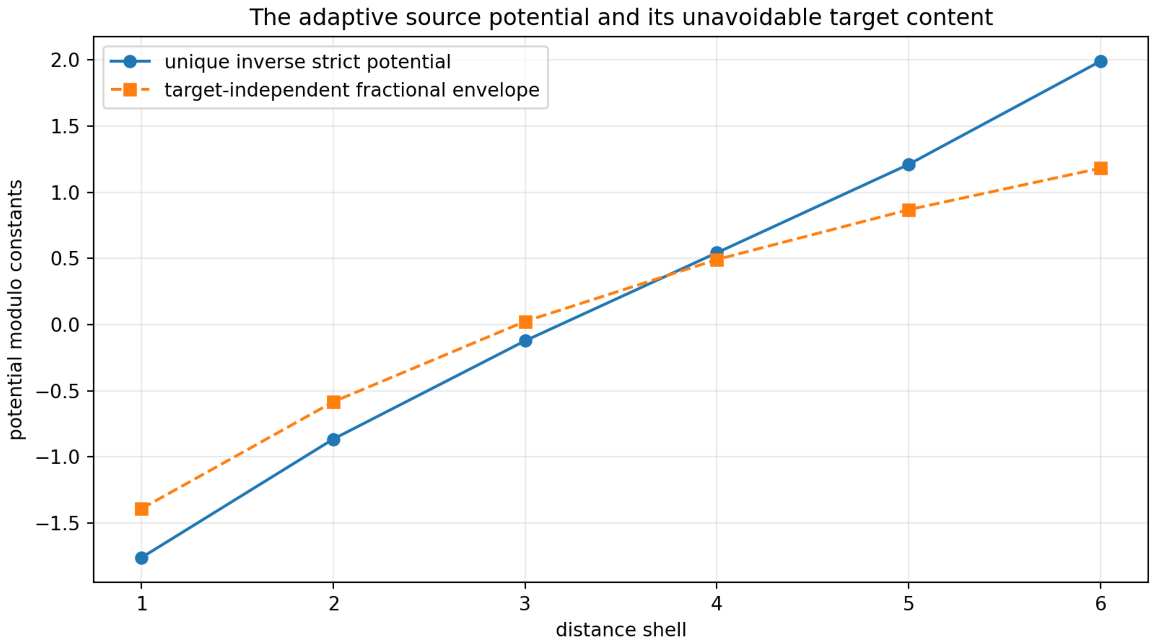

Figure 3.3: The unique inverse strict Fisher potential and the target-independent envelope potential.

3.4 Program 120 — Finite variational grammar

A finite target-free grammar was constructed from:

- five repository/canonical frequency candidates;
- four phase candidates;
- two beta candidates;
- eleven rational exponent candidates.

The total grammar contains

$$5 \times 4 \times 2 \times 11 = 440$$

actions. Each candidate was tested against strict distributions on both C_{12} and C_{24}.

No candidate is exact. The best mean divergence is

$$0.0023921978,$$

obtained at

$$(\omega, \phi, \beta, \eta) = \left(\frac{1}{12}, \frac{\pi}{4}, 1, \frac{9}{5}\right).$$

Adding the exact strict rationals makes the defect zero, but then the action carries the same four-parameter payload as direct storage of the kernel.

Negative result 120.1. Within the declared grammar, no exact target-independent action exists and target augmentation provides no description-length compression.

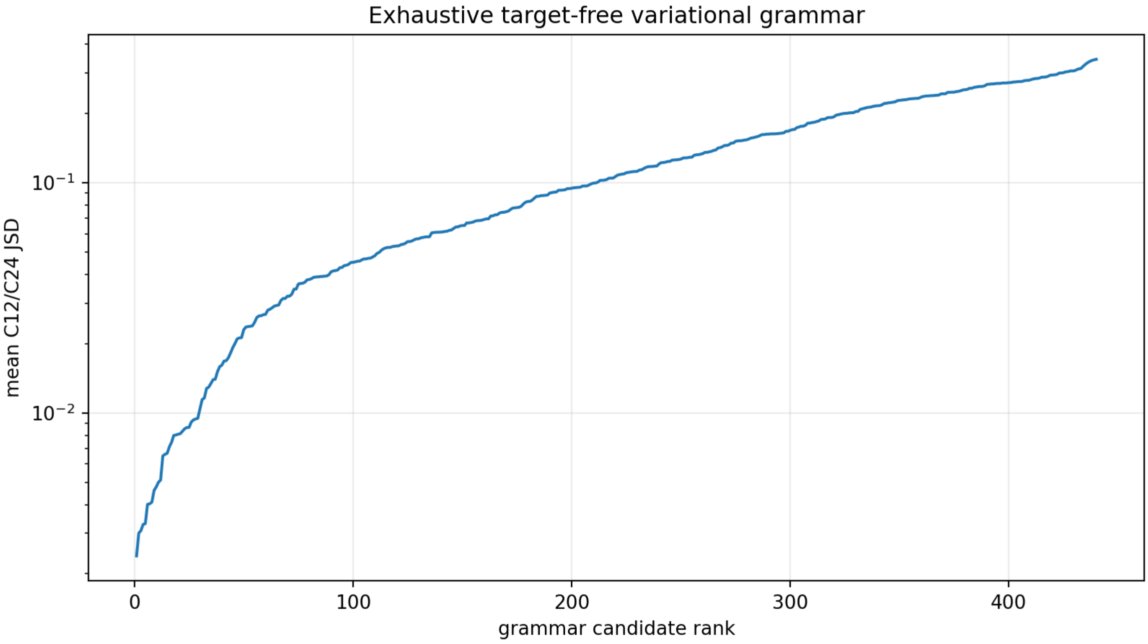

Figure 3.4: Exhaustive ranking of 440 target-free variational grammar candidates.

3.5 Program 121 — Hidden-Markov apparatus memory

The detector flip state $E_t \in \{0, 1\}$ is assigned transition matrix

$$\begin{pmatrix} 1-a & a \\ 1-b & b \end{pmatrix},$$

with

$$a = \epsilon(1-\rho), \qquad b = \epsilon + \rho(1-\epsilon).$$

For the test

$$\epsilon = 0.1, \qquad \rho = 0.8,$$

the channel is fit using 2,000 known-input calibration records. The fitted channel is frozen before held-out inference. A forward-algorithm likelihood then compares wave and diffusion hypotheses.

Synthetic held-out accuracy rises from 0.85 at five records to 0.9983 at one hundred records. The iid misspecified model is retained as a control; the HMM is not claimed to dominate it in every finite simulation, but it is the correct likelihood object for correlated apparatus memory.

Result 121.1 — [Constructed and synthetically validated]. A calibrated correlated-apparatus model can be integrated into the operational process without fitting on the scientific holdout.

No external evidence follows from synthetic data.

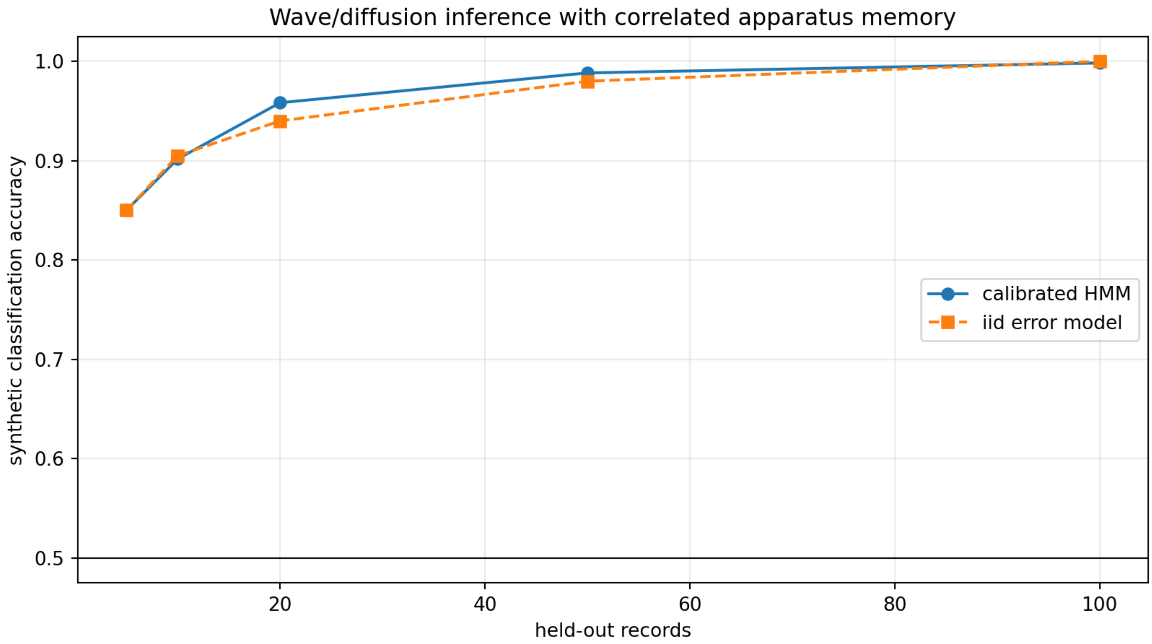

Figure 3.5: Synthetic held-out inference with calibrated hidden-Markov apparatus memory.

Chapter 4

Part IV — Programs 122–124: source object, damping theorem, and physical boundary

4.1 Program 122 — Z_{12} homological-character fibre object

4.1.1 Construction

Let

$$m_p : Z_{12} \to Z_{12}, \qquad m_p(x) = px.$$

For prime $p < 12$, the kernel contains $\gcd(p, 12)$ elements. The reduced zero-homology of that discrete kernel has dimension

$$\dim \widetilde{H}_0(\ker m_p) = \gcd(p, 12) - 1.$$

Let

$$\widehat{U(12)}_{\mathbb{R}} = \mathrm{Hom}(U(12), \{\pm 1\}).$$

It contains four real characters, three nontrivial. For a unit prime p, let $\mathcal{X}_p^-$ be the span of character basis vectors satisfying $\chi(p) = -1$. For a nonunit prime set this component to zero.

Define

$$\mathcal{F}_p = \widetilde{H}_0(\ker m_p) \oplus \mathcal{X}_p^-.$$

4.1.2 Exact dimensions

The fibres are:

p	$\dim \widetilde{H}_0(\ker m_p)$	$\dim \mathcal{X}_p^-$	$\dim \mathcal{F}_p$
2	1	0	1
3	2	0	2
5	0	2	2
7	0	2	2
11	0	2	2

Hence

$$\dim \mathcal{F} = (1, 2, 2, 2, 2).$$

No numerical coefficient is fitted: dimensions add because the object is a direct sum.

Theorem 122.1 — [Proven]. Given the Z_{12} multiplication system and its real unit-character dual, the displayed coefficient-free fibre construction yields exactly the P2938 vector.

4.1.3 Trace obstruction

The uniform coordinate trace gives

$$\tau_{\mathrm{unif}}(\dim \mathcal{F}) = \frac{1+2+2+2+2}{5} = \frac{9}{5}.$$

But a general probability trace w gives

$$\eta(w) = \sum_p w_p \dim \mathcal{F}_p = 2 - w_2.$$

Therefore

$$\eta = \frac{9}{5} \iff w_2 = \frac{1}{5}.$$

Obstruction 122.2 — [Proven]. The fibre object is canonical after Z_{12} is supplied, but the commutative algebra of five sectors admits many tracial states. Current strict data do not force the uniform trace or even the weaker condition $w_2 = 1/5$.

This sharpens the source problem from "why the vector?" to "why this trace?".

4.2 Program 123 — Conditional damping cocycle

4.2.1 Beta from multiplicativity

Let the nonlinear tail be

$$T(d) = \beta d^{\eta}.$$

Its multiplicativity defect factors as

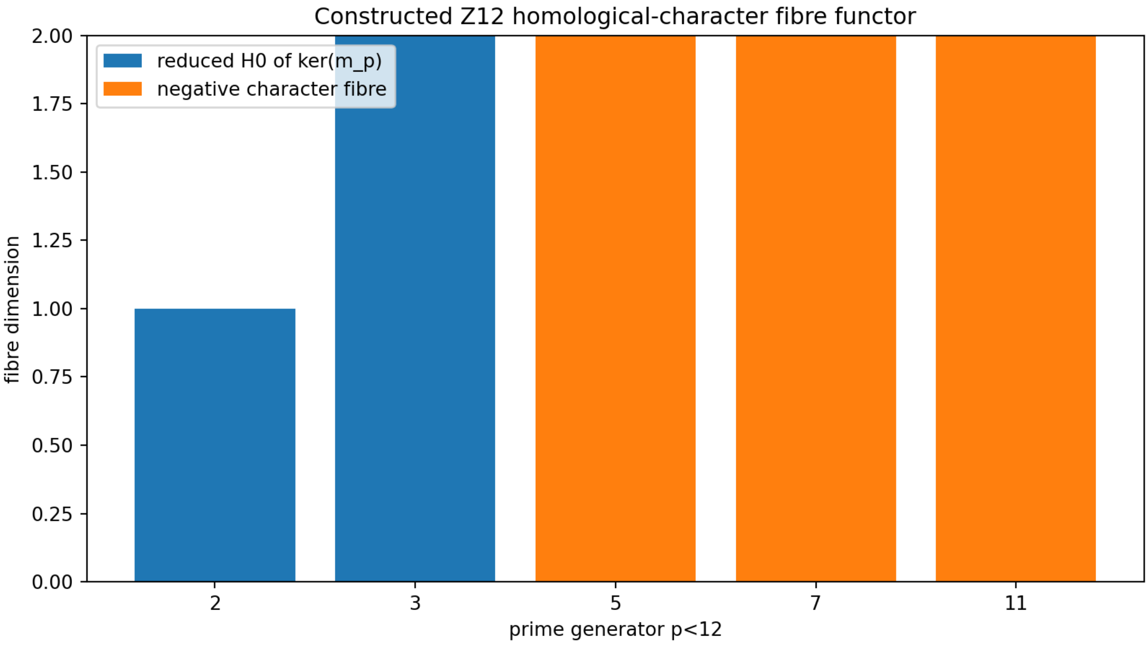

Figure 4.1: Homology and character contributions to the exact fibre dimension vector $(1, 2, 2, 2, 2)$.

$$T(ab) - T(a)T(b) = -a^{\eta}b^{\eta}\beta(\beta - 1).$$

For a nonzero positive tail,

$$T(ab) = T(a)T(b) \implies \beta = 1.$$

4.2.2 Eta and dyadic retention

Using the trace family from Program 122,

$$\eta = 2 - w_2.$$

The relative tail increment over the legacy degree one is

$$\delta = \eta - 1 = 1 - w_2.$$

The dyadic retention is therefore

$$r(w_2) = 2^{-\delta} = 2^{w_2 - 1}.$$

At $w_2 = 1/5$,

$$r = 2^{-4/5} = \exp(-\alpha_{\rm geo}/5).$$

The damping completion multiplier is

$$D_w(d) = \frac{1 + \beta_{\rm tors} d}{1 + d^{2-w_2}}.$$

At $w_2 = 1/5$ this maps the legacy linear damping envelope exactly to

$$\frac{1}{1+d^{9/5}}.$$

4.2.3 Minimal axioms and removal test

The conditional theorem uses three axioms.

1. **Fibre axiom:** the strict Z_{12} carrier is read through $\mathcal{F}_p$.
   - Remove it: the dimension vector disappears.
2. **Trace axiom:** $w_2 = 1/5$, for example through the uniform trace.
   - Remove it: $\eta = 2 - w_2$ remains continuously free.
3. **Multiplicative-tail axiom:** the nonzero T is a monoid character.
   - Remove it: positive β remains free.

Theorem 123.1 — [Conditional, proven]. These three axioms generate

$$(\eta, \beta) = \left(\frac{9}{5}, 1\right)$$

and the exact damping completion multiplier.

The first object has now been constructed. The second and third principles remain unexported strict source laws. No phase, frequency, amplitude, length unit, selector, or physical role follows.

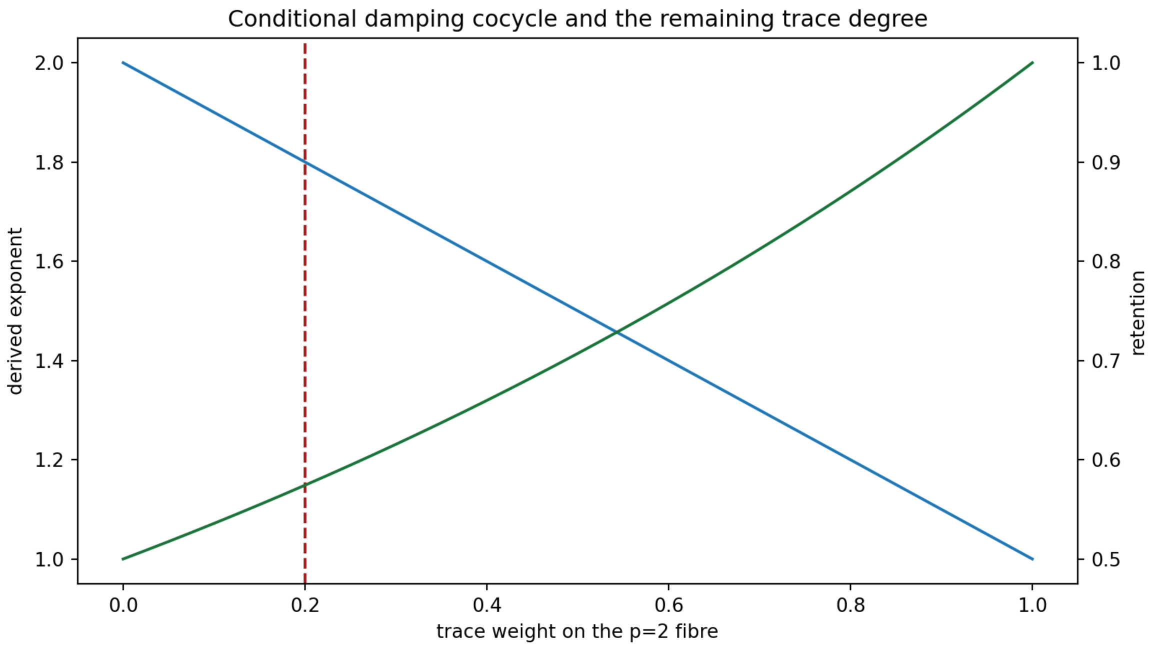

Figure 4.2: Exponent and dyadic retention as functions of the remaining trace weight.

4.3 Program 124 — Typed operational completion

The minimal operational object is extended to twelve typed fields:

1. state space;
2. state;

3. generator;
4. dynamics;
5. clock;
6. preparation;
7. instrument;
8. environment;
9. apparatus;
10. record;
11. calibration;
12. decision rule.

All twelve mathematical fields are present. This answers the earlier missing object problem at the level of a dimensionless operational theory.

Four physical inputs remain absent:

$$\ell_*, \qquad \tau_*, \qquad \hbar_*, \qquad \text{independent calibrated raw records.}$$

An immutable intake schema is frozen in FIN_Programs_113_124_External_Data_Intake.json. It requires hashes for raw records, preparation, instrument, graph/operator, clock calibration, apparatus calibration, environment/boundary description, and blind holdout split.

Result 124.1 — [Proven as a type-completion statement]. The operational mathematics can be made complete without resolving the dimensional bridge.

Boundary 124.2 — [Open]. No external data package satisfies the intake. The process is not experimentally validated.

Chapter 5

Part V — Synthesis and falsification

5.1 Constructed objects

This round constructs six previously missing objects.

Object	Construction	Result
Global Schur certificate	Exact rational factorization	Complete for all $m > 0$
Continuous parameter certificate	Infinite Hölder/integral bounds	Complete on declared box
Fractional limiting process	Stable-domain characteristic exponent	Complete dimensionlessly
Operational process	Ten/twelve typed components	Mathematically complete
Adaptive inverse potential	$V = -\log(q_*/r) + c$	Complete but target-equivalent
Z_{12} source fibre	Homology plus character negative space	Exact vector $(1, 2, 2, 2, 2)$
Damping cocycle	Trace exponent plus monoid character	Conditional $(9/5, 1)$

5.2 What remains genuinely missing

The constructions reduce the open frontier to smaller typed atoms.

1. **Trace source:** a strict reason for $w_2 = 1/5$.
2. **Compression law source:** a strict reason that the damping tail is the selected nonzero monoid character.
3. **Phase and frequency source:** no constructed object derives 743/4000 and 13/80.
4. **Amplitude bridge:** no role-safe theorem absorbs or transports α_{geo}.
5. **Selector:** all new fractional and trace objects remain inversion-even.
6. **Physical conversion:** $\ell_*, \tau_*, \hbar_*$ remain independent calibration data.
7. **Experiment:** no immutable external records are admitted.

5.3 Failed alternatives

5.3.1 Uniform trace as automatic

It fails because $\mathbb{C}^5$ has a simplex of tracial states. Uniform weight is a valid choice, not a theorem from commutativity alone.

5.3.2 Multiplicativity alone as eta source

It fails because every exponent satisfies

$$d^\eta e^\eta = (de)^\eta.$$

Multiplicativity fixes nonzero $\beta = 1$ but is eta-blind.

5.3.3 Inverse adaptive potential as explanation

It fails because the unique potential encodes the target exactly.

5.3.4 Finite action grammar

It fails because no target-free candidate is exact on both graph sizes.

5.3.5 Fractional dynamics as selector

It fails because $|q|^{4/5}$ is even under $q \mapsto -q$.

5.3.6 Type-complete experiment as physical evidence

It fails because a mathematical record channel is not an experimental record.

5.4 Current deepest interpretation

The deepest interpretation surviving this round is:

$$\boxed{\text{FIN strict lattice} \longrightarrow \text{symmetric 4/5-stable information process}}$$

together with a finite Z_{12} homological-character object whose dimension profile is exactly the strict exponent numerator package.

The relation is not yet inevitable because the normalized trace connecting the finite fibre dimensions to the tail exponent is not sourced. The current best candidate unification is therefore a **trace-coupled spectral information process**, not yet a physical field theory.

5.5 Confidence table

Statement	Status	Confidence
Global mass nonclosure	Proven	0.999
Continuous parameter-box nonclosure	Proven	0.995
Finite-q sub-one-percent certificate	Proven at 20 points	0.995
Symmetric 4/5-stable limit	Standard theorem applied	0.98
Fractional operational distinction	Proven finite model	0.999
Local/fractional crossover law	Proven	0.999
Unique inverse Fisher potential	Proven	0.999
Declared 440-action grammar has no exact source	Proven finite exhaustion	0.999
HMM apparatus object	Constructed	0.99
Fibre dimensions are $(1, 2, 2, 2, 2)$	Proven	0.999
Uniform trace is strictly selected	Open	0.10
Conditional damping gives $(9/5, 1)$	Proven conditionally	0.999
Complete legacy-to-strict bridge	Open	0.08
Physical validation	Open	0.00

Chapter 6

Part VI — Recommended Programs 125–137

6.1 Ranked next research programs

6.1.1 Program 125 — Uniform-trace source theorem or no-go

Probability of decisive result: 0.78
Priority: 1

Classify every state on the five-sector algebra compatible with the actual automorphism, tensor, and multiplication structure of $\mathcal{F}$. Prove either that these constraints force $w_2 = 1/5$ or that a nontrivial trace simplex survives.

6.1.2 Program 126 — Nadsoliton-to-fibre localizer

Probability: 0.55
Priority: 2

Construct a typed natural transformation from the strict nadsoliton carrier to

$$p \mapsto \widetilde{H}_0(\ker m_p) \oplus \mathcal{X}_p^- .$$

Test origin covariance, independence from a presentation of Z_{12}, and compatibility with existing P2938/P2958 provenance packets.

6.1.3 Program 127 — Effective Diophantine remainder

Probability: 0.62
Priority: 3

Derive a computable discrepancy bound for $743/(8000\pi)$ and combine it with the Fourier series of $|\cos|$. The deliverable is a continuous estimate

$$\left|1 - \widehat{p}(q) - C|q|^{4/5}\right| \le R(q).$$

6.1.4 Program 128 — Quantitative stable functional limit

Probability: 0.68
Priority: 4

Prove a rate in a probability metric for convergence to the stable process, including the oscillatory envelope remainder and truncation error.

6.1.5 Program 129 — Fractional wave dispersive theorem

Probability: 0.60
Priority: 5

Construct the full fractional wave group on a rigged Hilbert space, state the required UV domain, derive dispersive/stationary-phase bounds, and identify which detector observables remain finite without a cutoff.

6.1.6 Program 130 — Physical calibration completion

Probability: 0.40 without data; 0.85 with standards
Priority: 6

Attach independent $\ell_*, \tau_*, \hbar_*$ calibration classes to the typed operational object. Prove the dimensional rank and uncertainty propagation. Do not infer one calibration direction from dimensionless FIN records.

6.1.7 Program 131 — Nonparametric apparatus tomography

Probability: 0.70
Priority: 7

Replace the two-state HMM by a finite-memory process tensor learned only from calibration records. Derive identifiability, regularization, and robust wave/diffusion likelihood bounds.

6.1.8 Program 132 — Crossover RG phase diagram

Probability: 0.72
Priority: 8

Classify fixed points and crossover manifolds of $\kappa q^2 + \nu|q|^{4/5}$ under declared field normalization. Test whether any finite-scale local window can coexist with the infrared fractional limit.

6.1.9 Program 133 — Strict phase/frequency source object

Probability: 0.18
Priority: 9

Search for one new coefficient-free object producing $\omega = 743/4000$ and $\phi = 13/80$. It must not use these rationals in its definition or selection criterion. Stop after one bounded candidate class.

6.1.10 Program 134 — Role-safe amplitude bridge

Probability: 0.22
Priority: 10

Construct an amplitude normalization/absorption morphism explaining the passage from explicit α_{geo} in legacy to the unit-amplitude strict profile. Audit separately whether any legacy role survives.

6.1.11 Program 135 — Full damping bridge certificate

Probability: 0.35, conditional on Programs 125–126
Priority: 11

Install the trace and monoid character in an explicit legacy-to-strict damping map, prove uniqueness, and test held-out distances. Do not include phase, frequency, amplitude, or role transfer in the certificate.

6.1.12 Program 136 — State-dependent signed source

Probability: 0.15
Priority: 12

Construct one nonradial, inversion-odd signed state law coupled to the operational fractional process. Require nonzero value, reflection covariance, origin independence, stability, and a fixed torsor polarity. Otherwise preserve QW-2191.

6.1.13 Program 137 — Execute immutable external-data protocol

Probability: 0.00 without new records; 0.80 with admissible records
Priority: 13

Admit data only through the frozen Program-124 schema. Run the HMM/process likelihood on a blind holdout and report null results without retuning.

6.2 Stop rules

The next round must stop or issue a bounded no-go if:

1. a trace is declared uniform only because it gives 9/5;
2. the nadsoliton localizer assumes the P2938 vector;
3. phase or frequency candidates contain 743/4000 or 13/80 in their premises;
4. multiplicativity is reused as an eta selector;
5. physical units are inferred from a dimensionless trace;
6. a fractional even symbol is promoted to a chiral selector;
7. a synthetic HMM result is represented as experimental evidence;
8. a damping theorem is promoted to full bridge or role transfer.

Chapter 7

Reproducibility statement

Release 10.12 contains:

- the executable Programs 113–124 research suite;
- a machine-readable result record;
- 28 regression tests;
- eleven figures;
- a frozen external-data intake;
- this English source monograph;
- an archival English PDF;
- a Zenodo-ready release description;
- a checksum manifest.

No web crawler was used. No external experimental data were admitted.

Chapter 8

Final conclusion

The missing exponent object can now be written explicitly:

$$\boxed{\mathcal{F}_p = \widetilde{H}_0(\ker m_p) \oplus \mathcal{X}_p^-}$$

with

$$\dim \mathcal{F} = (1, 2, 2, 2, 2).$$

The theory has therefore advanced from an unexplained numerical vector to a canonical finite object. The remaining exponent obstruction is one scalar state on that object:

$$\eta(w) = 2 - w_2.$$

If the strict theory forces $w_2 = 1/5$ and a nonzero multiplicative compression tail, then

$$\eta = \frac{9}{5}, \qquad \beta = 1, \qquad r = 2^{-4/5} = e^{-\alpha_{\text{geo}}/5}$$

follow rigorously. At present those are conditional consequences, not strict exports.

The strongest surviving mathematical interpretation is thus a symmetric fractional spectral-information process coupled to a finite homological-character dimension object. The shortest next path is not another fit of the exponent. It is a theorem selecting — or proving the nonselectability of — the trace $w_2 = 1/5$.

Archival note

Machine-readable results are stored in `FIN_Programs_113_124_Constructive_Completion_Results.json`. The executable, twenty-eight tests, eleven figures, and frozen external-data intake reproduce the release.

Suggested citation:

Żuchowski, K. (2026). *FIN Programs 113–124: Constructive Completion Objects and Fractional Operational Physics* (FIN Research Monograph, Release 10.12; Version 1.0.0) [Preprint].